

Data Science

* Internal Question. *

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Q.1)* 1 - Mark Question (5 mark)

1) Give purpose of ^{Pre-}processing?

→ The purpose of preprocessing is it is responsible for maintain the quality of data. the phrase "garbage in, garbage out" is particularly applicable to such projects.

2) What is missing value?

→ Some values in the data may not be filled up for various reasons & hence it consider as missing values.

3) What is data Transformation?

→ Data Transform is the process of converting raw data into a format or structure that would be more suitable for data analysis.

4) What is data Discretization?

→ Data Discretization is characterized as a method of translating attribute values of continuous data into a finite set of intervals with minimal information loss.

5) Define data cleaning.

→ Data cleaning also known as data cleansing or scrubbing.

• Data cleaning is process of correcting or removing incorrect, incomplete or duplicate data within a dataset.

6) Define data Wrangling:

→ Data Wrangling is the task of converting data into a feasible format that is suitable for the consumption of the data.

7) What are different method for feature extraction

→ i) PCA (Principal Component Analysis)

ii) LDA (Linear Discriminant Analysis)

iii) GDA (Generalized Discriminant Analysis)

8) Why data reduction operation is important in data preprocessing.

→ data reduction is important phase in data preprocessing because it is carried out to reduce the unimportant or unwanted features from a dataset.

9) What is Data ~~cube~~ Cube.

→ A datacube is a multi-dimensional array of values. A data cube is generally used by easily interpret data.

10) differentiate Discrete & Continue Attribute

discrete att.	continue att.
i) it can be categorical or nominal	i) it can numerical or quantitative
ii) it represented using integers & strings	ii) it represented using real numbers
iii) eg - gender, color, product category	iii) eg - height, weight, temperature, price
iv) operation - ($=$, $\neq$, $\in$)	iv) operations - ($<$, $>$, $\leq$, $\geq$, $=$, $\neq$)

Q.2) 5-mark Que (Long Answer question)

1) difference between equal width & equal frequency binning.

equal width binning	equal frequency binning
i) It is easy to implement	i) it is more complex to implement
ii) it divide data into bins of equal size or width	ii) it divide data into equal bins with an equal no. of data point in each bin
iii) each bin has same range	iii) Bins may have different range
iv) it may not capture clusters in data	iv) it may not preserve original data distribution
v) eg - Age range 0-10, 11-20, 21-30, +---	v) eg - Age range 0-5, 6-15, 16-30, +---

2) Explain data cube Aggregation in detail.

→ it is a multidimensional aggregation which eases multidimensional Analysis.

• it present pre-computed & Summarized data which eases the data mining into fast Access.

• data cube is generally used to easily Interupt data.

• This figure has three dimensions - are → Time/year, brand & branch.

• data cube aggregation is a powerful tool for summarizing & visualizing large datasets.

• Aggregation involves applying aggregate function like (SUM, AVG, MAX, MIN).

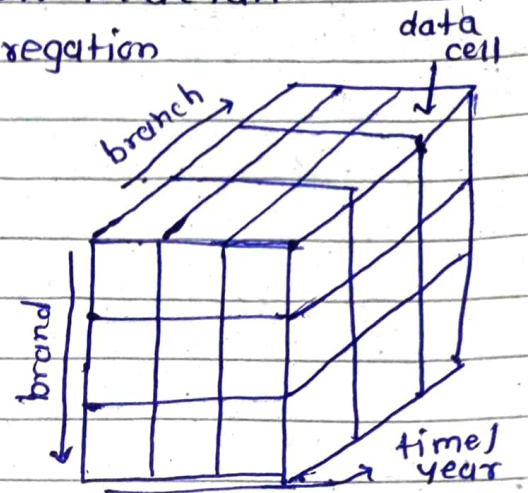


fig: Data cube aggregation

3) Explain Data Transformation Techniques / Strategies.

→ ① Rescaling :- Rescaling means transforming the data so that it fits within a specific scale, like 0-100 or 0-1

② Normalizing :- Normalization ensures that the attribute data to values which we are using in computations are not affected by trivial variations like height, width, scaling factors, orientations.

③ Binorizing :- it is the process of converting data to either 0 & 1 based on threshold value

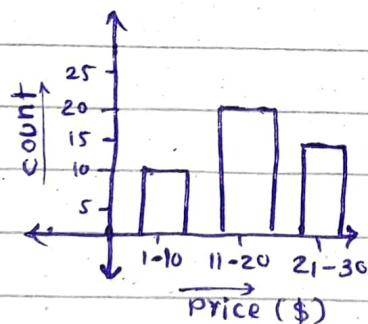
④ Standardizing :- it also called mean removal. it is another scaling technique where the values are centered around the mean with a unit standard deviation.

⑤ label encoding : label encoding is Process to used convert textual labels into numeric form in order to prepare it to be used in a machine-readable form.

④ What do you mean by discretization? Explain discretization by histogram analysis.

- discretization: discretization is characterized as a method of translating attribute values of continuous data into a finite set of intervals with minimal information loss.
- Histogram Analysis: Histogram analysis is the partitioning the values for an attribute into disjoint ranges called buckets.
- It is an unsupervised discretization technique because it does not use class information.
- To further reduce the data, it is common to have each bucket denote a continuous value range for given attribute.
- There are 2 partitions in Histogram.

① Equal Width histogram - In equal width histogram, the width of each bucket range is uniform.



② Equal Frequency Histogram:- In this histogram, the buckets are created so that, roughly, the frequency of each bucket is constant.

